

NATIONAL TRANSPORTATION SAFETY BOARD

Office of Research and Engineering
Materials Laboratory Division
Washington, D.C. 20594



March 20, 2007

MATERIALS LABORATORY STUDY

Report No. 07-018

A. ACCIDENT

Place : Boston, Massachusetts.
Date : July 10, 2006.
Vehicle : Connector Tunnel to Ted Williams tunnel.
NTSB No. : HWY06MH024.
Investigator : Mark Bagnard.

B. STUDY SUBJECT

Purging of a two part adhesive cartridge.

C. DETAILS OF THE STUDY

This study was conducted in order to demonstrate the effects of an air bubble in one of the two chambers of an epoxy cartridge.

Creation of Simulated Core-drilled Anchor holes

The simulated core-drilled anchor holes used for the testing were similar to those produced for the void formation study (Materials Laboratory Report No. 06-092) and consisted of clear acrylic tubing with an inside diameter of 0.75 inch that was cut into 5.5-inch lengths. A 0.25-inch high, 0.6-inch diameter disk of clear plastic was glued to the center of a 2 inch by 3 inch acrylic plate, and the acrylic plate was then glued onto one end of the acrylic tube such that the disk was approximately centered in the 0.75-inch diameter tube. For this study, a 5/8"-11 UNC nylon threaded rod was used as a substitute for the stainless steel anchor used in the tunnel, and the adhesive used was a Powers Fasteners two-part epoxy.

Testing

Powers Fasteners two-part epoxy is supplied in a cartridge that consists of two similarly sized chambers, one containing the resin, which is white, and the other containing the hardener, which is gray. A nozzle is built into the cartridge, centered between the two chambers, and allows material from each chamber to flow separately through it. The chamber nozzle is threaded to allow a mixing nozzle to be screwed onto it. The cartridges

are normally supplied in a box with the nozzle pointing upwards relative to the box markings.

The presence of an air bubble in one of the chambers would allow a volume of material from the other chamber equal to the volume of the bubble, to be injected into the mixing nozzle and not be subjected to the mixing process. Volumetric calculations indicated that a bubble of 0.1 inch in a cartridge chamber, nominally 1.6 inches diameter, would produce a volume of “unmixed” material equal to a cylinder $\frac{1}{2}$ inch deep in the test hole, nominally $\frac{3}{4}$ inch diameter (a nominal volume of 0.2 cubic inches¹). To reproduce the effect of such a bubble, a Powers Fasteners manual tool was used to dispense the adhesive (the manual tool is effectively a two barreled caulk gun with rams that move together). A 0.1 inch thick spacer was inserted and then removed from the front of a ram to provide the advanced flows required for the $\frac{1}{2}$ inch length of “unmixed” material in the test hole. As an undeterminable amount of the unmixed material would adhere to the walls and mixing elements of the mixing nozzle, and to the walls of the nozzle extension supplied with the cartridge, the nozzle extension was not used. For the same reason, the tests were then repeated using two 0.1 inch thick spacers² (for a total of 0.2 inch, or a nominal volume of 0.4 cubic inches of unmixed material).

The following steps were used to produce the test samples:

- (1) Insert a cartridge in the manual tool.
- (2) Purge the cartridge until material flows equally out of both chambers and note which color extrudes from which chamber.
- (3) Retract the rams on the manual tool.
- (4) Insert a 0.1 inch thick spacer on the forward face of the ram for the white resin chamber.
- (5) Install a new mixing nozzle.
- (6) Dispense the adhesive upward into a vertically oriented test hole identified as “A1”.
- (7) Retract the rams.
- (8) Remove the spacer.
- (9) Remove the mixing nozzle.
- (10) Install a new mixing nozzle.
- (11) Dispense the adhesive upward into a vertically oriented test hole identified as “B1”.

Repeat steps (3) to (11) using two spacers of 0.1 inch thickness to produce test holes identified as “A2” and “B2” respectively.

The first purging of the cartridge (Lot # 062650L1LC with a use by date of 09/2008) to achieve a balanced flow of resin and hardener was collected in a plastic cup and is illustrated in figure 1 with the white resin and the gray hardener identified. The amount of white resin extruded was significantly greater than the gray hardener, and a component of the hardener indicated by the red arrow had started to separate.

¹ A volume of 0.2 cubic inches is equivalent to a sphere (bubble) with a diameter of 0.65 inches.

² Actual thickness of the spacers was 0.104 inch and 0.107 inch.

Observation of the mixing nozzle during initial extrusion of the adhesive revealed that the color difference between the resin and the hardener was easily discernible. Figure 2 illustrates the mixing nozzle for sample "B1" with the hardener and the resin indicated. This sample simulated an air bubble in the resin chamber, allowing an advance flow of hardener. The flow has progressed approximately 3.5 inches in figure 2, and beyond the first mixing element the material appears generally to be a uniform gray color similar to the hardener.

Samples "A1", "A2" (with advanced white resin), "B1" and "B2" (with advanced gray hardener) were produced and allowed to cure for approximately 24 hours before being examined³.

To begin the examination process, the acrylic plates were removed from the test holes. At all four test holes, the disk remained in the hole as the acrylic plate was removed. The disks in test holes "A1" and "A2" remained firmly embedded in epoxy, and probing at both ends of these samples revealed that the epoxy had hardened (cured). The disks in the upper ends of test holes "B1" and "B2" were extracted easily from uncured epoxy at the top of the holes and streaks of uncured gray epoxy material dripped onto the ends of the tubes during the extraction process. Probing at the lower end of test holes "B1" and "B2" revealed that the epoxy at that location had hardened (cured). The upper ends of the four test holes are illustrated and identified in figure 3. On test holes "A1" and "A2" the disk is identified, and the yellow arrows indicate the gap, between the disk and the tube, which was used for the probing. On test holes "B1" and "B2" the upper end of the nylon threaded rod is identified, and the red arrows indicate the drips produced during the disk extraction. The unmarked black arrow on test hole "B2" indicates the upper edge of a void.

Further examination of the exterior of test holes "B1" and "B2" revealed areas of lighter coloration and rounded edges, indicating a thin layer of material on the inner surface of the tube, normally consistent with voids. The edges of the lighter colored areas were marked with a red dashed line and are illustrated in the center image in figure 4. The black arrows indicate voids adjacent to the upper end of the identified test hole, and the green arrow on test hole "B2" indicates a void adjacent to the void indicated by the black arrow. The yellow arrow on test hole "B2" indicates a longitudinally oriented void with an almost transparent portion located at the tip of the arrow. To illustrate the visual extent of the voids a sheet of transparent plastic was wrapped around each test hole, and the red dashed profile of the void was transferred onto the plastic sheet. Later, with additional markings (described later in this report) added to it, each plastic sheet was placed on a sheet of graph paper, with $\frac{1}{4}$ inch divisions, to produce the charts located on the left and right of the test holes illustrated in figure 4.

To determine the extent of the uncured epoxy, a stainless steel wire (0.036 inch diameter) was inserted into the upper end of each test hole several times, and at various angles, and the location where the wire was stopped was marked on the outer surface of

³ A chart on the cartridge indicates that with the base material at 75°F, the adhesive requires 24 hours to fully cure.

the test hole. These locations were also transferred to the plastic sheet and connected to produce a profile representing the uncured epoxy.

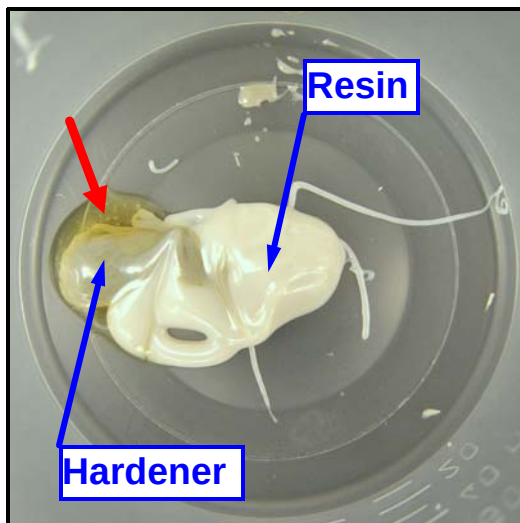
The left image in figure 4 illustrates the chart for test hole "B1", as indicated by the large blue arrow. The red line represents the shape of the void indicated by the black arrow on test hole "B1" in the center image. The black line represents the profile of the uncured epoxy and the zigzagged portion of the line indicates where the proximity of the threaded rod to the inner diameter of the test hole prevented further penetration of the wire.

The right image in figure 4 illustrates the chart for test hole "B2", as indicated by the large blue arrow. The red lines represent the shapes of the voids indicated by the black, green and yellow arrows on test hole "B2" in the center image. The black line represents the profile of the uncured epoxy and the zigzagged portion of the line indicates where the proximity of the threaded rod to the inner diameter of the test hole prevented further penetration of the wire.

In order to examine the surface of the epoxy in test holes "B1" and "B2", the nylon test anchor, and the epoxy, was pressed out of the acrylic tube 7 days after installation of the test anchors. Initial examination of the epoxy surface revealed areas that were tacky when touched, indicative of partially cured epoxy. The extent of the tacky areas was determined by touch, and the green dashed lines on the charts in figure 4 indicate their approximate boundaries. The extracted anchors, "B1" and "B2", are illustrated and identified in the center images of figure 5 with black arrows indicating the voids indicated by the black arrows in the center images of figure 4.

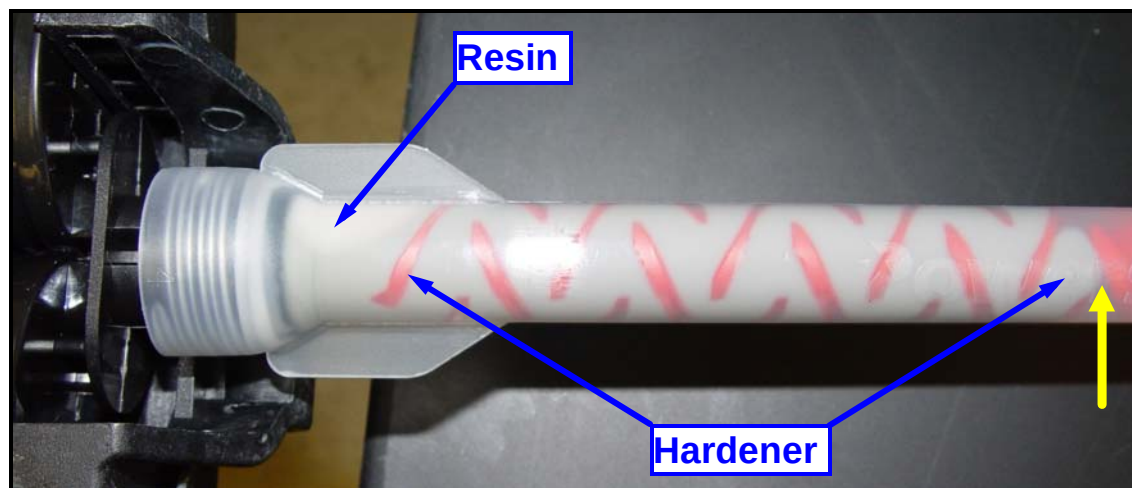
Examination of the test anchors revealed features that were similar to features observed on the anchors removed from the tunnel during post-accident investigation. The texture of the surface indicated by the green arrow on test anchor "B1" was similar to the texture of the surface observed on several tunnel anchors. The textured area was also located within the tacky area. Tunnel anchor "EB080-N-4/5/6-NE" is illustrated in the left image of figure 5 with a green arrow indicating a similarly textured surface that displayed a brown discoloration. Uncured epoxy similar to that indicated by the red arrow on test anchor "B2" was also found at the tip of several tunnel anchors. The right image in figure 5 shows tunnel anchor "EB081-S-10-S, and the red arrow in this image indicates uncured epoxy at the top of the anchor. Upward angling cracks similar to those noted on the smooth epoxy surface of test anchors "B1" and "B2" were also observed on several tunnel anchors, and examples are indicated by the blue arrows on tunnel anchor "EB081-S-10-S" in the right in figure 5. The edges of the cracks on the tunnel anchor displayed a brown discoloration also observed on the remaining surface of the anchor.

Derek Nash
Mechanical Engineer



ImageNo:0701A00724, Project No:2007010010

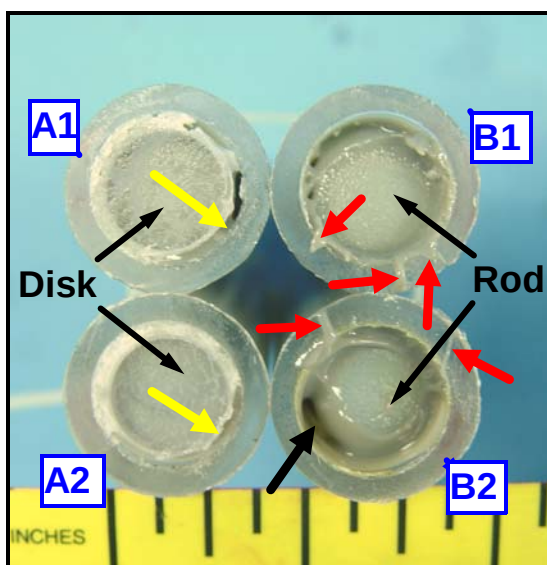
Figure 1. The initial resin and hardener that were extruded from the adhesive cartridge.



ImageNo:0701A00717, Project No:2007010010

2 in

Figure 2. The nozzle for trial "B1" at initial extrusion of the hardener (the gray colored material) and the resin (the white colored material).



ImageNo:0702A00066, Project No:2007010010

Figure 3. The upper ends of test holes "A1", "A2", "B1" and "B2" after removal of the plates, and attempts to remove the disks.

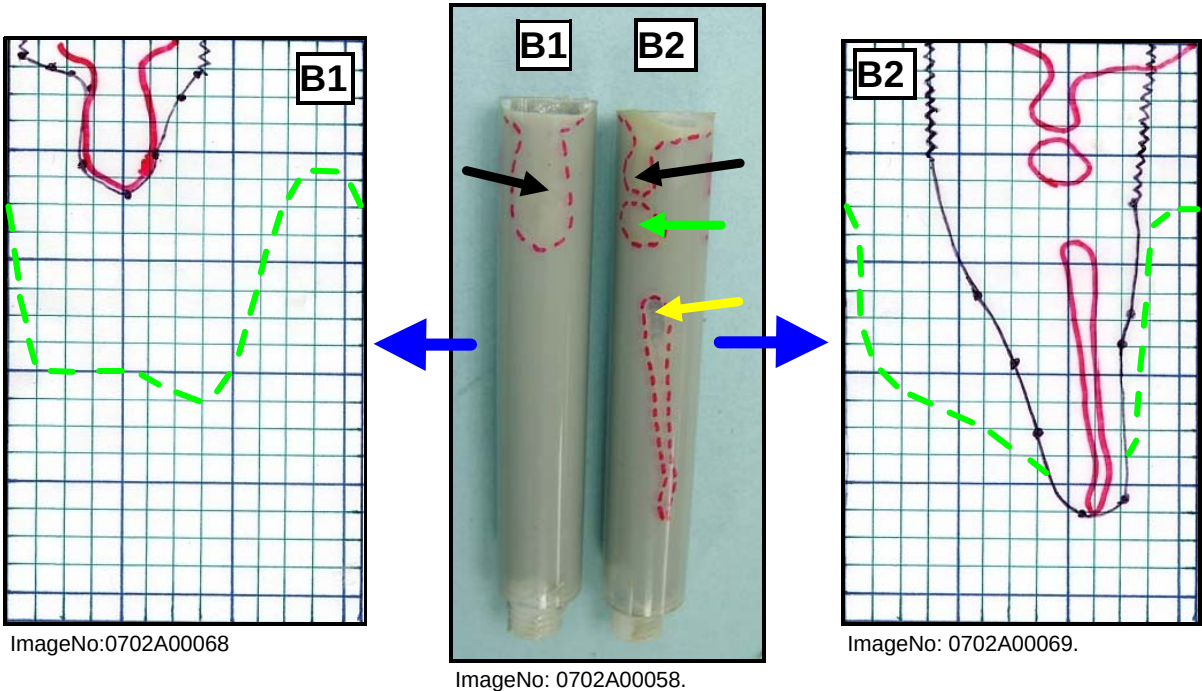


Figure 4. Test holes "B1" and "B2" (left and right respectively in the center image) with their respective charts as indicated by the large blue arrow.

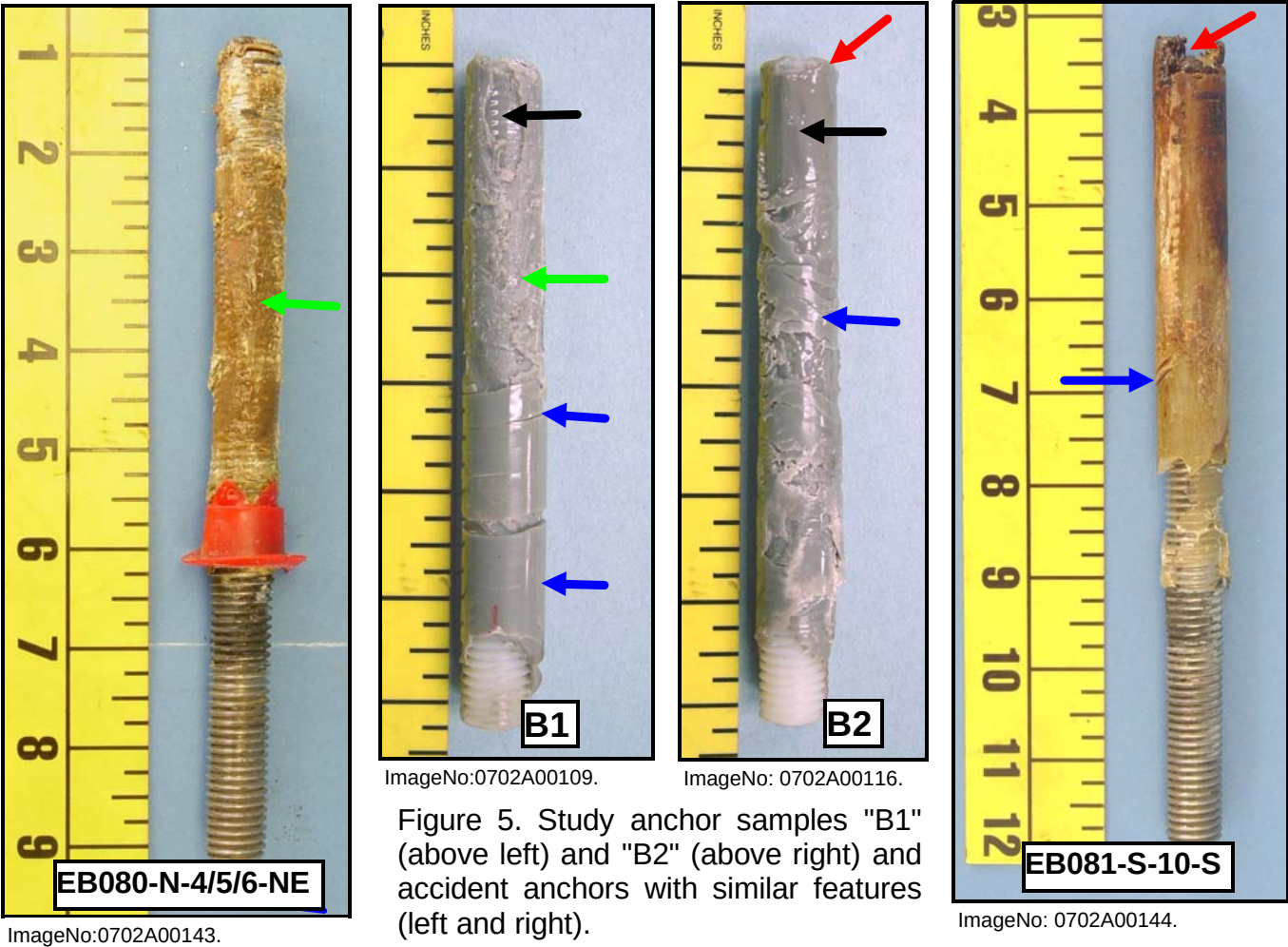


Figure 5. Study anchor samples "B1" (above left) and "B2" (above right) and accident anchors with similar features (left and right).